

ARM PrimeCell

COLOUR LCD CONTROLLER (PL110)

Errata Notice

ARM PrimeCell COLOUR LCD CONTROLLER PL110 Errata Notice

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1 ABOUT THIS DOCUMENT

1.1 Change control

1.1.1 Change history

This refers to the revision changes to the Errata document

Issue	Date	Change
1.0	12 th December 01	First Draft
2.0	7 th January 03	Second Draft
A	9 April 03	Updated for release r1p2-00ltd0

1.2 Errata List

This table summarises the errata notices detailed

Release: Release revision number affected

Section: Section in the document where the issues are described

Status: The status conditions are:

Open: Issue is current

Closed: Issue resolved in subsequent release

Severity: The severity conditions are:

A **Low** level of severity indicates that there is no direct impact to the silicon implementation. Examples are:

- No change to RTL required
- Documentation change
- Test pattern change
- Synthesis Scripts change

A **Medium** level of severity indicates that there is a minor impact to the silicon implementation, which can be overcome by modifications detailed in the errata notes. Examples are:

- A specific sequence of events not adhered to.
- Ambiguous or unexpected operation of the device.

A **High** level of severity indicates that there is a major impact to the silicon implementation, which cannot be overcome without modifications detailed in the errata notes. Examples are:

- Non conformance to the device specification.

Affected: Deliverables impacted by the issue

Description: Issue concerning Errata

Release	Section	Status	Severity	Affected	Description
1v0	2.1	Closed	Medium	RTL	CLCP can stop
1v1	3.1	Closed	Medium	RTL	Burst reads from registers return incorrect data
1v1	3.2	Closed	Medium	RTL	CLCP can stop
1v1	3.3	Closed	Low	Testbench	Incorrect data masking in test vectors
1v1	3.4	Closed	Low	RTL	Vector width mismatches
1v1	3.5	Closed	Low	RTL	Signal missing from sensitivity list
1v1	3.6	Closed	High	RTL	Watermark Level Inconsistency
r1p2-00ltd0	4.1	Open	Low	RTL	Timing violations may occur in high-performance systems

2 PL110 RELEASE 1V0 - ERRATA DETAILS

2.1 CLCP can stop

ARM reference D/E 139360

2.1.1 Summary

There are cases when the LCD Panel clock (CLCP) can be stopped, i.e. it freezes and sticks at 1 when it should be toggling, following a change in the PCD value in the Timing2 register.

2.1.2 Severity

Medium

In most real systems it is unlikely that the PCD would be changed, typically this would be set to a fixed value for a given panel. But, in some cases, a system may want to change PCD to change the screen refresh rate. For example, if changing the BPP to a higher value more data is required, it may be that the refresh rate must be lowered for there to be enough bandwidth. Or example 2, in a low power mode if the refresh rate was reduced then the rate of data transfer and hence power used is reduced. In some of these cases the clock generator may stop when PCD is changed.

2.1.3 Symptoms

Under certain circumstances the panel clock generator block, ClcdCPGen, can stop, i.e. stop clocking which stops the CLCP output from toggling. This occurs sometimes when the Panel Clock Divisor (PCD) is changed from an odd value to an even value. E.g. from 3 -> 2 or 1 -> 0. It will remain locked up while the PCD is even, e.g. 2 -> 0, but will recover when the PCD is returned to an odd value, e.g. 0 -> 1. It does not lock up every time, but this is not predictable.

2.1.4 Affected Items

ClcdCPGen

2.1.5 Short term work around

Do not change the PCD value if that is possible.

2.1.6 Proposed Corrective action

This issue has been fixed in release r1p2-00ltd0

3 PL110 RELEASE 1V1 - ERRATA DETAILS

3.1 Burst reads from registers return incorrect data.

ARM reference D/E 139165 and 110810

3.1.1 Summary

When a burst read is made from a group of registers, all the values returned will be that of the first register accessed. This is particularly visible when using MultiCE debug tools.

3.1.2 Severity

Medium

Only burst accesses to the registers are affected.

3.1.3 Symptoms

Debug tools may display incorrect register values, even when the register value is actually correct. Burst reads from multiple registers will return invalid results for all except the first register. Single reads from registers will return the correct values.

3.1.4 Affected Items

ClcdAhbSIVf.v(hd)

3.1.5 Short term work around

The registers can be viewed correctly for debug using the AXD window. Ensure software accesses registers using SINGLE transfers.

3.1.6 Proposed Corrective action

This issue has been fixed in release r1p2-00ltd0

3.2 CLCP can stop

ARM reference D/E 139360

3.2.1 Summary

There are cases when the LCD Panel clock (CLCP) can be stopped, i.e. it freezes and sticks at 1 when it should be toggling, following a change in the PCD value in the Timing2 register.

3.2.2 Severity

Medium

In most real systems it is unlikely that the PCD would be changed, typically this would be set to a fixed value for a given panel. But, in some cases, a system may want to change PCD to change the screen refresh rate. For example, if changing the BPP to a higher value more data is required, it may be that the refresh rate must be lowered for there to be enough bandwidth. Or example 2, in a low power mode if the refresh rate was reduced then the rate of data transfer and hence power used is reduced. In some of these cases the clock generator may stop when PCD is changed.

3.2.3 Symptoms

Under certain circumstances the panel clock generator block, ClcdCPGen, can stop, i.e. stop clocking which stops the CLCP output from toggling. This occurs sometimes when the Panel Clock Divisor (PCD) is changed from an odd value to an even value. E.g. from 3 -> 2 or 1 -> 0. It will remain locked up while the PCD is even, e.g. 2 -> 0, but will recover when the PCD is returned to an odd value, e.g. 0 -> 1. It does not lock up every time, but this is not predictable.

3.2.4 Affected Items

ClcdCPGen

3.2.5 Short term work around

Do not change the PCD value if that is possible.

3.2.6 Proposed Corrective action

This issue has been fixed in release r1p2-00ltd0

3.3 Incorrect data masking in test vectors

ARM reference D/E 141359

3.3.1 Summary

There is an error in one of the tests in that it has incorrect masking, and one bit is not included in a read/write test.

3.3.2 Severity

LOW

The RTL is unaffected.

3.3.3 Symptoms

In one of the test vector "IntrTest.c, in the "R_WTest" routine:

...

```
HSA(UpBase,NSEQ,SINGLE,OK,WRD);  
HSR(,DATA,,0xFFFFFFFFC,); <== (a)HSA(LpBase,NSEQ,SINGLE,OK,WRD);  
HSR(,DATA,,0xFFFFFFFFC,); <== (b)HSA(IntrEn,NSEQ,SINGLE,OK,WRD);  
HSR(,DATA & 0x0000001E,,0x0000001E,);  
HSA(CntrlReg,NSEQ,SINGLE,OK,WRD);  
HSR(,DATA & 0x001BFFF,,0x000BFFF,); <== (c)
```

...

In location c), the test is in error. Bit 16 was added as an enhancement to the CLCD controller late in the development of the peripheral and the test was not updated.

3.3.4 Affected Items

IntrTest.c

3.3.5 Short term work around

A modification needs to be made to the test code:

For the test labelled (c), change the mask from

0x0000BFFF to 0x0001BFFF

3.3.6 Proposed Corrective action

This issue has been fixed in release r1p2-00ltd0

3.4 Vector Width Mismatches

ARM reference D/E 110609 and 110567

3.4.1 Summary

There are some vector width mismatches in some of the verilog files in the design. It is unlikely that these will cause problems that result in incorrect operation of the device.

3.4.2 Severity

Low

These do not affect the functionality of the device

3.4.3 Symptoms

Functionality is not affected.

3.4.4 Affected Items

ClcdAhbMasterIf.v

ClcdDMAFifo.v

ClcdOutMux.v

ClcdPalette.v

ClcdFormat.v

ClcdFifoCntl.v

3.4.5 Short Term Workaround

If necessary, the files can be checked using a linting tool, and the vector widths corrected.

3.4.6 Proposed Corrective action

This issue has been fixed in release r1p2-00ltd0

3.5 Signals Missing From Sensitivity List

ARM reference D/E 110568

3.5.1 Summary

A signal is missing from the sensitivity list in the file design. This could result in a latch being inferred, which would affect the operation of the design or result in mismatches between simulation and actual behaviour.

3.5.2 Severity

Low

Any problems are likely to have a minor impact on the actual operation of the device.

3.5.3 Symptoms

The interaction between CLLE and LcdEn may be incorrect.

3.5.4 Affected Items

ClcdTiming.v

3.5.5 Short Term Workaround

Line 706 needs to be changed accordingly to:

always @(LEStartSet or LEEEn or LEWait or CLLE or LEDel or LECCount or LcdEn)

3.5.6 Proposed Corrective action

This issue has been fixed in release r1p2-00ltd0

3.6 Watermark Level Inconsistency

ARM reference D/E 217617

3.6.1 Summary

There is a bug in the DMA FIFO logic of the LCD controller which causes some data corruption in the CLCDC. It is possible for a burst access to be generated which requests more data than can be stored in the FIFO. The extra data will be discarded, resulting in the data for the remainder of the frame being offset.

3.6.2 Severity

High

Transient display corruption can occur.

3.6.3 Symptoms

Data loss can occur which results in incorrect display data being output.

3.6.4 Affected Items

ClcdDMAFifo.v/vhd

3.6.5 Short Term Workaround

None available.

3.6.6 Proposed Corrective action

This issue has been fixed in release r1p2-00ltd0

4 PL110 RELEASE R1P2-00LTD0 - ERRATA DETAILS

4.1 Timing violations may occur in high-performance systems

ARM reference D/E 202414

4.1.1 Summary

Reads from several registers which are in the CLCDCLK domain are not correctly synchronised to the HCLK domain. This may result in setup/hold violations in high-speed simulations when a register is written and read back immediately. The register values themselves are correctly synchronised and are not affected. If the read occurs some time after the write, then the stable, correct value will be sampled.

4.1.2 Severity

LOW

Software is unlikely to access the peripheral in a way which will show this problem.

4.1.3 Symptoms

A write to LCDTiming0-3 or LCDControl, followed immediately by a read from that register may result in the old value being read back, or a transient incorrect value being sampled. If sufficient time is allowed for the register to be updated in the CLCDCLK domain, and this value to propagate back to the HCLK domain, then the correct value will be seen.

4.1.4 Affected Items

RTL

4.1.5 Short Term Workaround

This problem can only be corrected by modifying the interface to registers which are located in the CLCDCLK domain. Either the read values should be double synchronised, or the registers shadowed in the HCLK domain.

4.1.6 Proposed Corrective action

This issue may be addressed in a future release of PL110